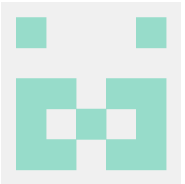


# The Calendars - [Cal.diy](https://github.com/CSCI-435-SE/cal.diy) - <https://github.com/CSCI-435-SE/cal.diy>

|                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Demir Beynam                                                                      | Zach Bowden                                                                       | Dylan Han                                                                         | Julissa Hernandez                                                                  | Joshua Ko                                                                           | Zachary Short                                                                       |
| dbbeynam                                                                          | zachbowden                                                                        | dchan01-wm                                                                        | julissaehp                                                                         | kojoshuay                                                                           | zach-short                                                                          |
|  |  |  |  |  |  |

## Sprint Backlog and Overview

This sprint, the team set out to deliver all 16 issues across five areas: UI & interaction (including the availability grid), core event management, a favorites system, notification infrastructure, and auth/account improvements. The scope was ambitious but well-defined from the start, spanning both user-facing features and backend foundations.

Scope remained largely stable throughout — no major features were added or cut. However, several issues proved more complex than initially estimated. What appeared as single, contained issues revealed deeper implementation work once underway, requiring them to be broken into sub-issues with multiple PRs. This was most notable in **[areas to confirm — e.g. the availability grid drag interaction, the favorites backend]**. The team adapted by decomposing the work rather than deferring it, which kept quality high but extended the timeline on those items.

[Sprint 1 Milestone](https://github.com/CSCI-435-SE/cal.diy/milestone/1) - <https://github.com/CSCI-435-SE/cal.diy/milestone/1>

Team decisions: When determining story points, one of our rules of thumb was that there is, in general, a hierarchy of complexity in changes: purely UI changes ~ purely schema changes < UI and schema changes. Also, our general guideline for determining the scope of an issue was that between 4-6 was a medium issue.

An example of our Story Point document:

#### Issue Story Points

- Website setting to force 2fa for privileged or non-privileged users #3
  - <https://github.com/CSCI-435-SE/cal.diy/issues/3>
  - 7
  - Hardest part is existing users to be prompted to have 2FA
    - Josh and Julissa
- Ability to set event theme #5
  - <https://github.com/CSCI-435-SE/cal.diy/issues/5>
  - 4
  - Setting color should not be too hard; will require changing the schema to remember what color is for what event
- Ability to set a time, not just date for being "out of office" #6
  - <https://github.com/CSCI-435-SE/cal.diy/issues/6>
  - 5
  - Will have to update the UI - create a new menu/field for day/time
  - Will also have to update the schema to include OOO day/time
- Warn user before leaving event edit page with unsaved changes #9
  - <https://github.com/CSCI-435-SE/cal.diy/issues/9>
  - 4
  - Will not have to adjust database; UI change; will have to listen for moving off the page (going to a different menu)
- Require a reason when a booking is rescheduled #11
  - <https://github.com/CSCI-435-SE/cal.diy/issues/11>

| Title                                                                               | Owner                                                                                             | Story Points | Scope  | Rationale                                                                                                                                                                           | Status                                                                                |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <a href="#">Implement Drag-to-Select Interaction on the Availability Grid - #70</a> | dchan01-wm<br>   | 5            | medium | Must create the logic to handle clicking / dragging / touching the screen and determine which cells must be selected/deselected. Must store this data somewhere; no schema changes. |    |
| <a href="#">Build the Weekly Availability Grid Component - #69</a>                  | dchan01-wm<br>   | 4            | medium | Must modify the UI to create a new calendar component. Does not deal with clicking/dragging logic yet, but must set up selection/deselection for #70. No changes to the schema      |    |
| <a href="#">Ability to set event theme - #5</a>                                     | zach-short<br> | 4            | medium | The UI change for setting color should not be hard; will require a database change to store the color per event                                                                     | WIP                                                                                   |
| <a href="#">Ability to set a time, not just date for being "out of office" - #6</a> | zach-short<br> | 5            | medium | Will have to update the UI with new a new menu for day/time; will have to update the schema to include Out of Office day/time                                                       |  |
| <a href="#">Warn user before leaving event edit page with unsaved changes - #9</a>  | zach-short<br> | 4            | medium | Purely a UI change, listening for clicking off the edit page (whether going to another page or leaving site); NO schema change required                                             |  |

|                                                                                                 |                                                                                                   |                                                                  |        |                                                                                                                                                                                                                               |                                                                                       |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <a href="#">Add a favorite button to event types to favorite/pin most frequent events - #15</a> | kojoshuay<br>    | ORIGINAL Y 5<br><br>SPLIT into two subissues:<br><br>#75 and #76 | medium | Requires an addition to the UI - the favorites button. Also requires a change to the schema to track which events are favorited                                                                                               |    |
| <a href="#">Add a rebook feature sub issue 1</a>                                                | Julissaehp<br>   | 2                                                                | low    | Reusing the existing booking flow keeps the feature simple and allows users to create a new booking without changing the original booking.                                                                                    |    |
| <a href="#">Allow users to set a custom default landing page after login - #19</a>              | julissaehp<br>  | 3                                                                | medium | Small schema change: per user, store the preferred landing page. Small logic change to set the initial page to load                                                                                                           |    |
| <a href="#">Reminder Preference Controls &amp; Schema - #65</a>                                 | zachbowden<br> | 5                                                                | medium | Building upon the simple 24 hour email reminder. With this change, there is the added complication of multiple time selections.                                                                                               |  |
| <a href="#">Add a rebook feature sub issue 2</a>                                                | julissaehp<br> | 5                                                                | medium | required creating a new People view, organizing previous attendees, calculating meeting counts and dates, and connecting the Book Again feature. It involved both frontend and backend changes, making it moderately complex. |  |

|                                                                                                        |                                                                                                  |   |        |                                                                                                                                                                                                                                                 |                                                                                       |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <a href="#">Background Scheduling &amp; Email Delivery - #66</a>                                       | zachbowden<br>  | 4 | medium | A fairly simple email reminder that gets sent out 24 hours before an event is set to start for both the host and attendee. Several logical issues such as figuring out if emails were to be sent out if events were moved had to be considered. |    |
| <a href="#">Persist user information during account creation when choosing alternative email - #27</a> | zachbowden<br>  | 4 | low    | No UI change; autopopulates existing fields. Should require no or small schema changes to persist the user information                                                                                                                          |    |
| <a href="#">Add database support for favorited event types - #75</a>                                   | kojoshuay<br>  | 4 | medium | Required Prisma schema modeling and repository methods following DI patterns                                                                                                                                                                    |   |
| <a href="#">Add favorite event types section to the Event Types dashboard - #76</a>                    | kojoshuay<br> | 6 | medium | Required full stack integration - building UI components, writing new query and mutation procedures for the tRPC layer, and bridging domain services                                                                                            |  |

## Requirements & design

For the most part, the specs were written and approved by another teammate before any coding started. There were a couple instances where we sent an approval message in Zulip, but forgot to add it as a Github issue, but those were remedied afterwards.

- kojoshuay

- The most important design choice I made was storing events user favorited as a separate table, as opposed to directly modifying EventType records or affecting teams. Allowed for users to have their own events that they favorited. Also, avoids pointer collision bugs that would come from using a backup column that stores the position of the event before favoriting.
- zach-short
  - #6 (out-of-office times, PR #85): The main design choice was to treat an out-of-office entry with times as a busy interval, subtracted from availability the same way calendar events are, instead of teaching the per-day "away" map about time ranges. Whole-day entries keep the existing path, and the two are told apart by the stored start and end, so no schema change or migration was needed. The cost is that a partial day shows no "away" badge on the booking page.
  - #9 (unsaved-changes warning, PR #78): The main design choice was one capture-phase click listener on the page while the form has unsaved edits, instead of a guard in each navigation component. It catches every link, including links added later, and needs no private Next.js API. The cost is that the browser Back button and Cmd+K navigation are not guarded.
    - Both specs have a user story, a scenario, acceptance criteria, Out of Scope and Open Questions. The open questions were answered in comments on #6 and #9 before code was written. Each PR has a Design section with the alternatives considered.

## Completed issues

| Issue Title                                                                                            | Issue Owner | Issue Creator | PRs                                                                  | PR Details                                                                        | Changes                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------|-------------|---------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Background Scheduling &amp; Email Delivery - #66</a>                                       | zachbowden  | kojoshuay     | <a href="#">feat: add 24 hour email reminders - #79</a>              | Author: <b>zachbowden</b><br>Reviewer: <b>dchan01-wm</b><br>Status: <b>Merged</b> | Add an email notification sent to both the host and attendee when the current time is 24 hours away from the event start time.                        |
| <a href="#">Persist user information during account creation when choosing alternative email - #27</a> | zachbowden  | dchan01-wm    | <a href="#">feat: persist user information on email change - #74</a> | Author: <b>zachbowden</b><br>Reviewer: <b>kojoshuay</b><br>Status: <b>Merged</b>  | Add the ability for the user creation form to be auto-populated with previously entered details if the user selects "use a different email" when they |

|                                                                                     |            |            |                                                                                              |                                                                                   |                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------|------------|------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                     |            |            |                                                                                              |                                                                                   | are at the email verification page.                                                                                                                                                                               |
| <a href="#">Add database support for favorited event types - #75</a>                | kojoshuay  | julissaehp | <a href="#">feat: 75 add database support for favorited event types - #77</a>                | Author: <b>kojoshuay</b><br>reviewer: <b>dchan01-wm</b><br>status: <b>Merged</b>  | Added the EventTypeFavorite join model to the Prisma schema with database migration. Implemented repository methods for querying, creating, and deleting user favorites. Also added unit/integration tests.       |
| <a href="#">Add favorite event types section to the Event Types dashboard - #76</a> | kojoshuay  | julissaehp | <a href="#">feat: 76 add favorite event types section to the event types dashboard - #80</a> | Author: <b>kojoshuay</b><br>Reviewer: <b>zachbowden</b><br>Status: <b>Merged</b>  | Added a "Favorites" section on the dashboard w/ star toggles for each event. Implemented tRPC procedures for querying and toggling favorite statuses to connect frontend with domain services and the repository. |
| <a href="#">Ability to set a time, not just date for being "out of office" - #6</a> | zach-short | zachbowden | <a href="#">feat: set a leave time and a return time for out of office - #85</a>             | Author: <b>zach-short</b><br>Reviewer: <b>zachbowden</b><br>Status: <b>Merged</b> | Out-of-office entries can have a start time and an end time. The booking page blocks only that window, and the rest of the day stays bookable.                                                                    |
| <a href="#">Warn user before leaving event edit page with unsaved</a>               | zach-short | kojoshuay  | <a href="#">feat: warn before leaving event</a>                                              | Author: <b>zach-short,</b>                                                        | Leaving the event type editor with unsaved edits                                                                                                                                                                  |

|                              |  |  |                                                      |                                                               |                                                                                                       |
|------------------------------|--|--|------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <a href="#">changes (#9)</a> |  |  | <a href="#">edit page with unsaved changes (#78)</a> | Reviewer:<br><b>kojoshuay</b><br><br>Status:<br><b>Merged</b> | opens a dialog to save, discard or stay. Closing or reloading the tab shows the browser's own prompt. |
|------------------------------|--|--|------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|

## Test strategy

The most common strategy for testing was writing unit tests. Most of the changes we made were fairly isolated, both in terms of loose coupling and how many components the code interacted with, so we could individually test sections of code. Manual testing was generally used for the UI changes. There is only so much you can do programatically to assert that a UI element looks and works right. For instance, when creating the calendar for selecting availability, we can write unit tests to assert that if a given time is selected, it should appear a different color, but the user experience and *seeing* if the changes work is best done manually. Asserting that the element is X pixels wide is fine, but *seeing* that it is what you expected (aligns with the requirements set out in the issue) is another thing.

## AI tool usage

- zachbowden: Used Claude Code for feature implementation, human verification for every line generated and manual testing of features in a development server.
  - <https://github.com/CSCI-435-SE/cal.diy/tree/main/ai-logs/sprint1/zachbowden>
- kojoshuay: Used Claude Code for planning and feature implementation. Making sure to verify the plan and every line generated. Features were also tested manually.
  - <https://github.com/CSCI-435-SE/cal.diy/tree/main/ai-logs/sprint1/kojoshuay>
- dchan01-wm: Used Claude Web for understanding issues and the code. Used Claude Code for understanding the code, implementation planning, test creation, code review, explanations of others' code, explanations of code it created. 6 sessions
  - [cal.diy/ai-logs/sprint1/dchan01-wm at main · CSCI-435-SE/cal.diy](#)
  - Notable patterns: spent extensive time asking the AI to explain the code it generated and to explain code others generated (code review). Last sprint, the code changes were so small that I could easily understand what the code did. Now, with my unfamiliarity with React/TS and the large scope of the changes, AI was useful to explain how different systems in the code work (e.g. the form.schedule field that stores availability information) and how the created code relates to existing systems.
- zach-short: Used Claude Code (Opus 5.5 for #9 and for my reviews of #61 and #81, Fable 5.1 for #6) to read the existing code, propose options for each open decision, implement the chosen design with tests, and walk each feature in a browser against the

dev server. I made the product decisions before any code was written and reviewed each diff before committing. Four logged sessions, linked from comments on #6, #9, #11 and #35.

- Julissaehp: Used claude code for issues 19, 87,88, I allowed claude to create changes, didn't use it for testing like i did last time. It was also used to read existing code base, and help look at all options for best outcome for each issue and implementation. All tests were run manually. I had three ai log sessions
  - <https://github.com/CSCI-435-SE/cal.diy/tree/main/ai-logs/sprint1/julissaehp>

## Release

- Tag name and link to the Sprint 1 release:  
<https://github.com/CSCI-435-SE/cal.diy/releases/tag/v6.2.0-csci435-s1>

## Risks and retrospective

- kojoshuay
  - Implementing the feature of a user being able to favorite events was a lot more complicated than I initially thought. This is because of the codebase's architecture standards: database access must go through repository classes, business logic must live in services, and both are wired through dependency injection. I also had to follow the existing tRPC conventions of a separate schema, handler, and route entry for an endpoint.
- dchan01-wm
  - I think I used Claude very effectively this Sprint 1. I spent a lot of time asking Claude to explain its design decisions and code. In doing so, we were able to refine the features to be implemented, and my manual questioning helped point out errors (Claude pointed out an error centered on code not running with an if block, but further questioning revealed that the problem was dealt with elsewhere!). With the approach of going back and forth to explain what I intended and to gain better understanding, I think Claude performed better; with my first PR, I believe it only have to fix one or two bugs, which it caught from our investigations (most of the other "bug" fixes were fixing biome linter problems). On the other hand, this in-depth approach really slowed down the process. I think for sprint 2 I can investigate using Claude to help me better automate my workflow and prompts I use frequently (e.g. so I don't have to manually write out prompts for getting Claude to explain how the code works).
- Julissaehp
  - Using Claude was helpful for understanding the codebase, planning my implementation, and writing tests, but there were also some risks because AI-generated code can make unnecessary changes or misunderstand parts of the project. For example, while working on the Book Again feature, Claude helped me reuse the existing booking flow and create tests, but I still had to

manually test the feature to make sure the attendee information was prefilled correctly and that the original booking was not changed. Claude also helped with the People view by organizing previous attendees and connecting the Book Again functionality, but I still needed to review the code and check the test results because some tests in the full test suite failed for unrelated parts of the project. Overall, using Claude made development faster and helped me work through unfamiliar parts of the codebase, but I learned that AI should be used as a tool and that I still need to review, test, and verify the code myself.

## Sprint 2 plan

- In the process of implementing Issue #32, we realized we could add further functionality to the calendar. For instance, we could add long-pressing vs a quick movement to differentiate between selecting/deselecting and scrolling through the calendar. We can also add functionality for selecting across days (currently we can only select availability across a given day). The selection code here could be reused in other places where dates are inputted. For instance, we could add a similar draggable calendar to the Out of Office menu, or in the event creation menu when specifying the availability of a given event.
- Another area the team can work on is enforcing two-factor authentication (2FA). Since 2FA is already implemented in the application, the goal would be to make sure certain users are required to use it. Issue #67 would require administrators to enable 2FA, while Issue #68 would require it for all users. Since both issues use the existing 2FA system and involve similar enforcement logic, the work from one issue could potentially be reused for the other.
- Overall just getting to the smaller issue that in the long run help when big features are being added in to make the user experience smoother.